From "Truth Topology" to Readout Bottlenecks: Negative Replication on GSM8K and a Positive Representation–Decoder Dissociation in Procedural Micro-Worlds

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Abstract

We report a multi-stage study of hidden-state geometry for correctness prediction in language-model reasoning. The initial hypothesis was that global topological summaries (especially persistent-homology H_0/H_1 features) would provide a robust correctness signal. That hypothesis failed under controlled checks: dynamic and windowed H_0 features were weak or degenerate and did not outperform simple baselines. In a separate fixed-decoding GSM8K branch, we found a stronger operational effect: non-convergence (token-cap termination) was the dominant predictor of error. We then moved to a procedurally generated language micro-world with nonce lexicons, held-out templates, and exact truth conditions over `True`/`False`/`Unknown`. Across Qwen and Gemma families, decoder behavior systematically under-expressed `Unknown`, yet verdict-region hidden-state probes recovered substantial `Unknown` signal on held-out worlds. This representation–decoder dissociation persisted under prompt-path controls, constrained label decoding, base-vs-instruct checks, verdict-step logit analysis, and layer sweeps. The main positive result is therefore not a global "truth topology" scalar, but a replicated internal-semantic/readout mismatch for non-entailment.

1 Scope and Claim

This repository started as a "topology of truth" project. The final evidence supports a narrower and stronger claim:

> In a held-out procedural semantic task, non-entailment (`Unknown`) is represented in hidden states but systematically under-realized by decoder outputs.

The paper is therefore a combined report of:

1. negative results for global topology summaries on math traces,
2. a convergence-failure result in fixed-decoding GSM8K runs, and
3. a replicated representation–decoder dissociation in a controlled micro-world.

2 Experimental Trajectory

2.1 Phase A: Global Topology on GSM8K (Negative)

We first tested static and dynamic topology features on small paired runs (Qwen3.5-0.8B/2B). The dynamic H_0 pilot on non-capped 2B traces (9 samples) showed weak separation: `h0_entropy_final AUC=0.55`, `peak_h0_entropy AUC=0.55`, `auc_h0_entropy_per_token AUC=0.50`, and `max_delta_h0 AUC=0.60`.

The subsequent windowed correction test showed that fixed-window topology remained non-useful:

- controls-only AUC: 0.60,
- corrected-topology-only AUC: 0.20,
- controls+corrected-topology AUC: 0.20.

These outputs are in `artifacts/qwen35_2b_window_corrected/`.

2.2 Phase B: Fixed-Config Within-Question GSM8K (Convergence Result)

We ran Qwen3.5-2B with fixed decoding (`temperature=0.7`, `top_p=0.95`, `max_new_tokens=384`) on a within-question batch ($n = 180$ total samples). The strongest effect was termination mode:

- capped: 87 (75 wrong / 12 correct),
- EOS: 93 (17 wrong / 76 correct),
- AUC(cap $\rightarrow$ wrong): 0.839,
- odds ratio (wrong given cap): 26.40.

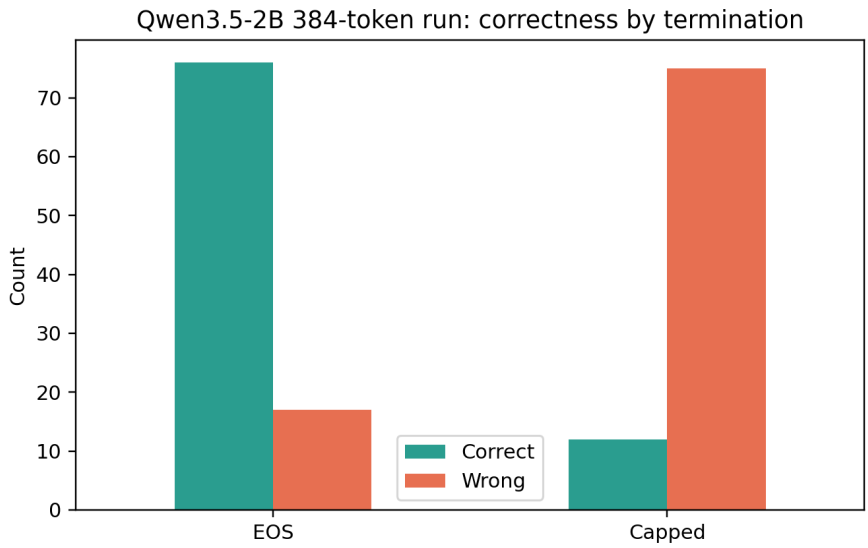

Figure 1: Fixed-config Qwen3.5-2B run: correctness split by termination mode.

For sensitivity, we reran all originally capped samples ($n = 87$) at `max_new_tokens=640` with matched seeds/IDs:

- 59 reached EOS, 28 remained capped,
- among 75 originally capped wrong: 27 flipped wrong→correct, 48 stayed wrong.

Interpretation: both truncation and deeper non-convergence contribute.

2.3 Phase C: Procedural Micro-World Semantics (Main Positive Result)

To reduce benchmark contamination and isolate semantic reasoning, we built a synthetic micro-world dataset:

- latent structured worlds with nonce vocabularies,
- held-out lexicons/templates across splits,
- exact tri-label truth status: `True`/`False`/`Unknown`,
- 72 examples per world (9 propositions $\times$ 8 paraphrases),
- 20 train worlds + 20 eval worlds used in the main model sweeps.

3 Core Micro-World Results

3.1 Decoder Unknown Collapse vs Hidden Unknown Recoverability

Table 1 compares decoder behavior to verdict-token probe recovery on held-out eval worlds.

Table 1: Decoder vs verdict-token probe on eval micro-worlds ($n = 1440$ per model).

Model	Decoder Acc.	Decoder Unknown Recall	Probe Unknown Recall
Qwen3.5-2B	0.436	0.000	0.738
Qwen3.5-4B (no-think)	0.478	0.000	0.129
Gemma-3-4B-it (no-think)	0.397	0.013	0.563

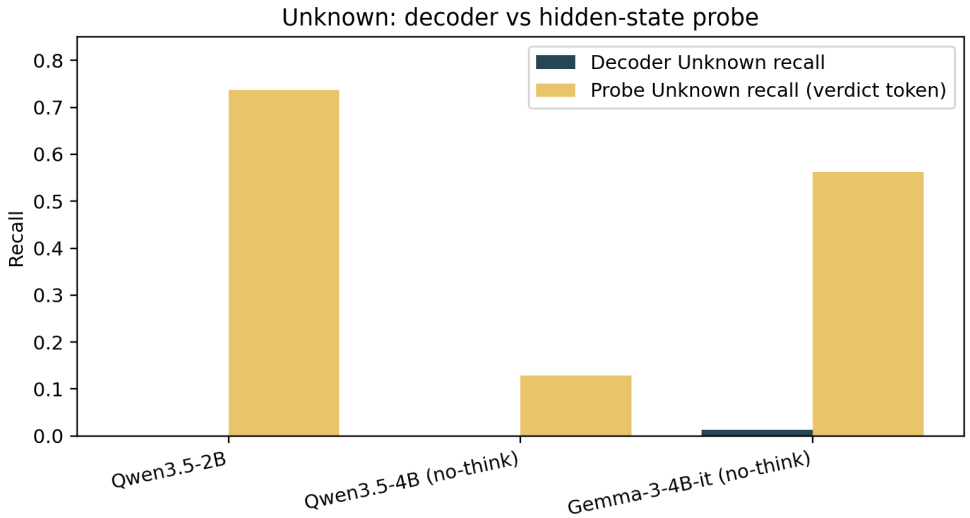

Figure 2: Unknown recall: decoder outputs vs verdict-token probe. Hidden-state signal substantially exceeds emitted-label signal.

3.2 World-Level Geometry Consistency

Within-world geometry tests compare same-label vs different-label verdict-state distances. For Qwen3.5-2B (19 valid eval worlds) and Gemma-3-4B-it (20 worlds), all tested state keys had positive sign-test rates:

$$\text{different-label distance} > \text{same-label distance}$$

in 19/19 and 20/20 worlds, respectively.

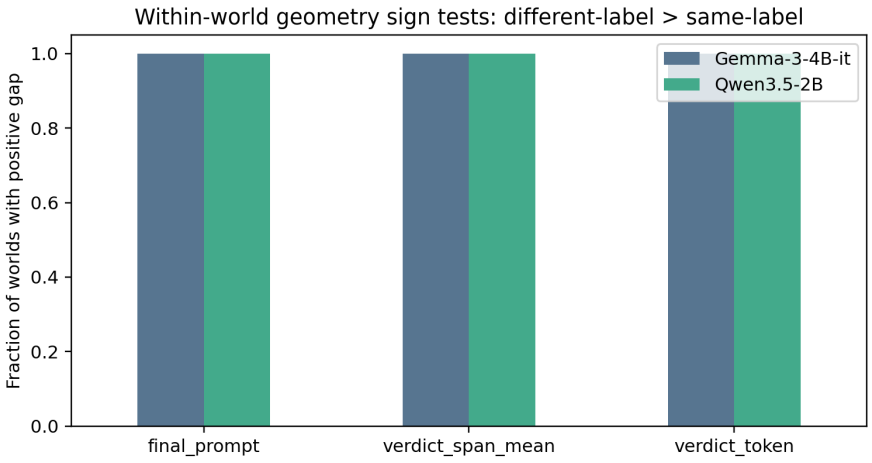

Figure 3: Fraction of worlds with positive same-vs-different label distance gap.

3.3 Mechanistic Controls: Prompt Path, Constrained Decode, Base vs Instruct

The dissociation survived multiple controls:

- raw-prompt path for Gemma-it still showed decoder/probe mismatch,
- constrained {`True`,`False`,`Unknown`} decoding did not reliably repair Unknown emission,
- initial Gemma-base parse failures under mismatched prompt format were repaired using a base-specific label format,
- after parse-confound repair, Gemma-base still had decoder Unknown recall = 0 while probes recovered Unknown.

3.4 Verdict-Step Logit Analysis

On gold-`Unknown` decoder failures, verdict-step logits under-ranked `Unknown`:

- Gemma-3-4B-it: mean $P(\texttt{Unknown}) = 0.192$, mean log-prob margin -1.293,
- Gemma-3-4B-pt (basefmt): mean $P(\texttt{Unknown}) = 0.177$, mean log-prob margin -1.045.

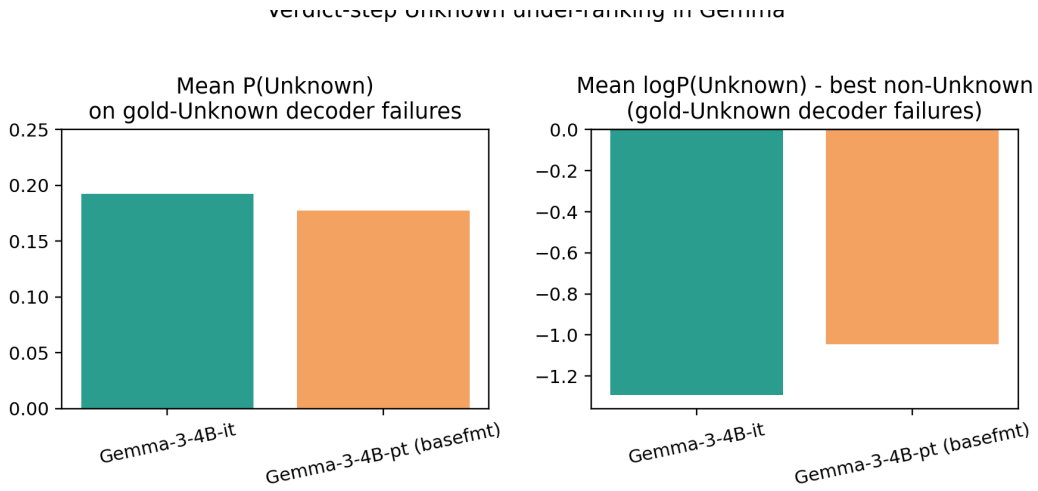

Figure 4: Verdict-step evidence of Unknown under-ranking on gold-Unknown failures.

3.5 Layer Sweep

Layer-wise probes showed strong internal Unknown recoverability in both Gemma variants, with model-dependent layer locations. Best Unknown recall by state family:

- Gemma-it: prompt_last 0.779 (layer 1), verdict_token 0.742 (layer 10),
- Gemma-pt (basefmt): prompt_last 0.825 (layer 29), verdict_token 0.733 (layer 28).

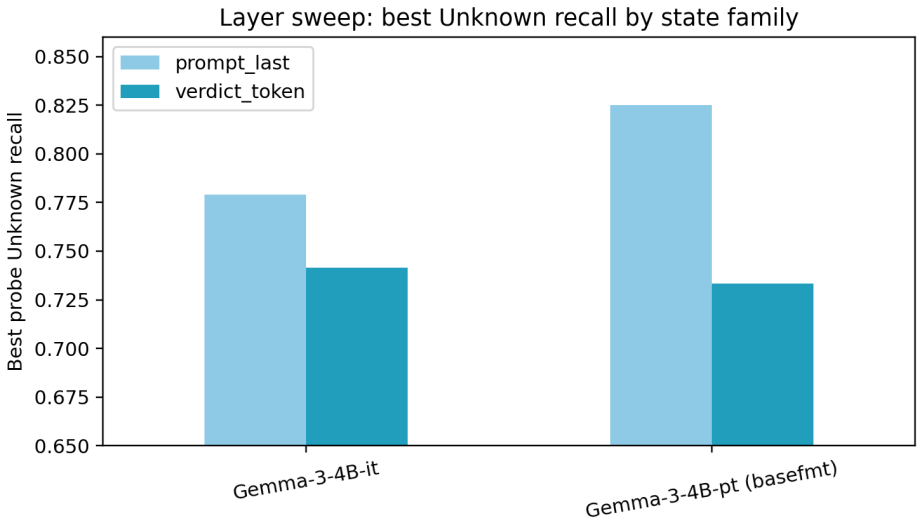

Figure 5: Best Unknown recall from layer sweeps by state family.

4 Interpretation

The combined evidence supports a readout bottleneck interpretation:

1. semantic non-entailment is encoded internally,
2. decoder outputs under-express that class,
3. the mismatch is replicated across model families and controls.

We do **not** claim a universal global topology scalar for correctness. That formulation was not supported by the GSM8K/global-H_0 branch.

5 Contributions

1. A complete negative replication record for global-trace topology correctness signals.
2. A fixed-decoding convergence analysis showing cap-linked non-convergence as a dominant error mode.
3. A procedural, held-out semantic benchmark with exact tri-label truth status.
4. A replicated representation–decoder dissociation for non-entailment across model families.
5. Mechanistic triangulation via constrained decoding, prompt-path controls, logit-level analysis, and layer sweeps.

6 Reproducibility and Artifacts

All produced outputs are committed under `artifacts/` and `outputs/` in this repository. Key analysis entry points:

- convergence branch: `artifacts/qwen35_2b_within_question/analysis/`,
- cap sensitivity: `artifacts/qwen35_2b_cap_sensitivity_640/analysis/`,
- micro-world comparisons: `artifacts/micro_world_v1/comparison_*.csv`,
- layer sweep: `artifacts/micro_world_v1/layer_sweep_*`,
- label-logit analysis: `artifacts/micro_world_v1/label_logits_*`.

Paper figures are generated deterministically from committed CSV files via:

```
python3 paper/scripts/make_figures.py
```

7 Limitations

- We studied small-to-mid model sizes only; larger-scale behavior is open.
- The micro-world is controlled and compositional, but still synthetic.
- Probe recoverability does not, by itself, prove causal use by the decoder.

8 Conclusion

The original "truth topology" objective was too broad for the observed evidence. The final, validated contribution is a replicated representation–decoder dissociation: non-entailment is internally represented but systematically under-surfaced at output time. This reframes the problem from discovering a single topology-of-truth scalar to studying how semantic uncertainty is lost between internal representation and decoder readout.